

FIRAS FRANÇOIS ABED EL NABI

TECH LEAD & FULL-STACK ENGINEER | JAVA | ANGULAR | DDD/ SOFTWARE ARCHITECTURE

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SUMMARY

Tech Lead with 10+ years building complex enterprise software, currently leading a team of 5 at Adobis Group (Grenoble). PhD in Computational Material Science turned software engineer - I bring research-grade problem decomposition to messy business domains, and I own architecture rather than only implementing it: DDD and Hexagonal/Ports & Adapters across Java/Spring back-ends and Angular/TypeScript front-ends.

Outside work I build and self-host my own AI-assisted development tooling on Linux VPS infrastructure, and hold a deliberately critical line on where AI belongs in an engineering workflow.

EXPERIENCE

Adobis Group - Grenoble, France
Tech Lead & Full-Stack Developer

August 2021 - Present

Leadership & architecture

- Lead a team of 5 developers: sprint planning, code review, mentoring, and onboarding
- Made new-joiner ramp-up repeatable rather than ad-hoc - built and ran the onboarding programme, 4 engineers onboarded to date
- Made new back-end services testable independently of database and framework by rolling out DDD + Hexagonal Architecture
- Decoupled front-end logic from framework concerns with a Ports & Adapters architecture over reactive Angular components
- Enabled 5 applications to build and release independently from one codebase - Nx monorepo with micro-frontend delivery
- Removed the AngularJS dependency, end-of-life since 2021, by migrating 7 legacy modules to Angular over 18 months
- Ran Scrum ceremonies (standups, poker planning, retrospectives, backlog management); gathered requirements directly from customers and produced proofs of concept

Delivery

- Let formula authors write and check dataframe expressions in-editor before Spark executes them - a custom business DSL on a Lezer parser with CodeMirror
- Made automation processes reconfigurable without a code change - replaced hand-coded workflow definitions with a visual JointJS editor supporting per-task configuration
- Gave every front-end module one source of UI truth instead of per-module re-implementation - a Storybook-documented design system of 23 components (15 atomic, 8 integration)
- Centralised authentication for hundreds of users across 5 tenants behind one Keycloak OAuth2/OIDC flow, including a custom token mapper

Technical environment: Java 11/17/21, Spring Boot/MVC/Security/Data, Hibernate, Redis, Scala, Spark, Mockito | Angular, TypeScript, NgRx, RxJS, ImmutableJS, Immer, Angular Material, JointJS, AngularJS, Backbone, Nx | Keycloak, OAuth2, OIDC, Docker | Storybook, Jest, Karma, Jasmine, Mocha

Independent Engineering & Self-Hosting - Personal projects

2024 - Present

- Remove repetitive scaffolding from day-to-day development by shipping internal tooling - code-generation scripts, test scaffolding, and documentation generators - with AI handling a substantial share of the implementation
- Integrate AI-assisted development into daily engineering workflow (Claude Code, custom agents, MCP integrations, automated pipelines) for code generation, test scaffolding, prototyping, and architectural exploration
- Hold a deliberately critical line on where AI belongs: high value for boilerplate, tests, and rapid exploration; human ownership retained for architecture, domain modelling, and production-critical decisions
- Own the full production path for personal and professional applications on a Hostinger VPS - deployment, domains, DNS, SSL/TLS, and server configuration
- Keep releases repeatable and low-ceremony by automating deployment and maintenance through scripting and CI/CD workflows

Technical environment: Linux, Docker, CI/CD pipelines, Bash, Python | Claude Code, MCP, custom agents

Client: EATON - Intelligent Power Manager (IPM2)

- Built web interfaces for energy consumption monitoring across UPS-backed data centres and facilities, including Eaton's own Grenoble-Montbonnot site
- Kept monitoring views consistent under streaming updates with NgRx state management and RxJS reactive flows in an object-oriented Angular/TypeScript architecture
- Coordinated work packages across back-end and firmware teams

Technical environment: Angular, TypeScript, AngularJS, NgRx, RxJS, ImmutableJS, Immer, Angular Material

Client: Schneider Electric - Wiser Energy

- Built interfaces for energy consumption monitoring and photovoltaic production tracking in residential installations, delivered as internal and end-user applications across EMEA
- Delivered home automation for Bluetooth Low Energy actuators - light bulbs and shutters - through a web application deployed to Android and iOS
- Shipped two product front-ends: WiserEnergy (AngularJS) and WiserOne (Angular)

Technical environment: Java 8 | React, Angular, TypeScript, AngularJS, NodeJS, ThreeJS | Ionic Cordova

WISEO Technologies - Grenoble, France

March 2017 - June 2018

Full-Stack Consultant

Client: Politecnico di Milano - SPARK (Spatial Augmented Reality as a Key for co-creativity)

- Full-stack developer on an EU research project projecting 3D designs onto 3D-printed physical models in real time
- Built back-end services on Spring MVC/Data with Apache Storm for stream processing
- Built the front-end rendering pipeline in AngularJS and ThreeJS
- Integrated heterogeneous components over message-oriented middleware (ActiveMQ), bridging Java and C# subsystems

Technical environment: Spring MVC, Spring Data, Apache Storm, JPA/Hibernate, SQL, Elasticsearch | AngularJS, ThreeJS, Gradle, Gulp | C#, ActiveMQ

SELECTED PROJECTS

Spark formula DSL editor (Adobis) - IDE-style editor for authoring dataframe formulas executed by Spark; Lezer parser + CodeMirror

Visual workflow engine (Adobis) - JointJS editor for automation workflows with per-task custom configuration

Design system library (Adobis) - 23 shared components (15 atomic, 8 integration), Storybook-documented, adopted across every module

Nx micro-frontend architecture (Adobis) - Monorepo spanning 5 apps on Ports & Adapters; carried the AngularJS to Angular migration of 7 modules

AI-assisted dev tooling (personal) - Code generation, test scaffolding, and documentation generators, self-hosted on a Hostinger VPS

TECHNICAL SKILLS

Expert - Java (11/17/21), Spring Boot, Spring MVC, Spring Security, Spring Data, Hibernate | Angular, TypeScript, RxJS, NgRx |DDD, Hexagonal Architecture, Ports & Adapters

Working - PostgreSQL, MySQL, Redis, Elasticsearch | Keycloak, OAuth2, OIDC | Nx, Docker, GitLab CI, Jenkins | JUnit, Mockito, Jest, Karma, Jasmine, Mocha, Storybook | NodeJS, React, NextJS, NestJS | JointJS, ThreeJS, D3JS | Python, Bash

Familiar - Scala, Spark | MongoDB, Neo4j | C#, ActiveMQ | Vite, Webpack, Rollup, Gradle

EDUCATION

Java Software Developer Bootcamp | M2I, Grenoble, France 2016 - 2017

PhD, Computational Material Science | Université de Poitiers / Institut PPRIME, France 2012 - 2016

MSc, Particle Physics | Université Grenoble Alpes (Joseph Fourier), France 2009 - 2012

LANGUAGES

French - Fluent | English - Fluent | Arabic - Native | German - Intermediate